

Pramay Bhatpahri

PhD Scholar, Aerospace Engineering
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Research Profile

Doctoral researcher in computational granular mechanics, developing discrete element method (DEM) simulations and constitutive models for small-strain elasticity of dense spherical granular assemblies. Research interests include granular micromechanics, contact mechanics, particle breakage, wave propagation, shock absorption, scientific machine learning, and topological descriptors.

Education

Ph.D., Aerospace Engineering 1 Aug 2019 – Present
Indian Institute of Science, Bengaluru
Advisor: Prof. S. Gopalakrishnan.
Thesis: *Small-Strain Constitutive Models for Dense Granular Materials*.

M.Tech., Machine Design (Honours) 2014 – 2016
National Institute of Technology, Raipur
CPI: 8.35.
Thesis: *Fatigue and Fracture Behavior of API X65 Line Pipe Steel*; collaboration with Tata Steel R&D, Jamshedpur.

B.E., Mechanical Engineering (First Class with Honours) 2009 – 2013
CCEM, Raipur; Chhattisgarh Swami Vivekanand Technical University, Bhilai
CPI: 8.63.

Doctoral Research Experience

PhD Researcher Aug 2019 – Present
Department of Aerospace Engineering, IISc Bengaluru

- Developed YADE triaxial simulation workflows for dense spherical packings under isotropic confinement and deviatoric loading; compared Hertz–Mindlin and Cundall–Strack contact laws.
- Extracted early-loading tangent elastic moduli and Poisson's ratio using elastic-window detection and consistency checks.
- Calibrated separable stiffness laws across particle Young's modulus, confining stress, and void ratio using trust-region nonlinear least squares and physical admissibility constraints.
- Compared empirical models with neural-network and multi-layer perceptron surrogates; evaluated models using AIC/BIC, grouped cross-validation, leave-one-design-out diagnostics, bootstrap analysis, and permutation tests.
- Analysed particle-breakage effects on stiffness scaling, pressure exponents, void-ratio dependence, and separability.
- Computed 36 fabric and force-anisotropy invariants, including contact-fabric, normal-force, tangential-force, force-support, and cross-tensor descriptors, to relate microstructure to macroscopic stiffness.

Technical Skills

Programming	Python, MATLAB, C, shell scripting; Fortran for Abaqus UMAT workflows.
Simulation	DEM: YADE, LAMMPS. FEM: Abaqus, ANSYS.
CAD	SOLIDWORKS, CATIA V5, AutoCAD.
Scientific Tools	NumPy, Pandas, SciPy, scikit-learn, Matplotlib; ParaView, VTK.
Research Tools	LaTeX, Git/GitHub, VS Code, Overleaf; scientific plotting and reproducible analysis workflows.

Publications

Published Journal and Conference Papers

1. **B. Pramay**, Anay Mohan Shembekar, and S. Gopalakrishnan, "Early-loading tangent modulus maps for dense spherical DEM assemblies across contact laws and particle-modulus regimes," *Particuology*, 2026. Available online 3 October 2026.
   DOI: 10.1016/j.partic.2026.09.014 | Article.

2. **Pramay B.** and S. Gopalakrishnan, "Separable surrogates for DEM-derived elastic moduli of Hertz–Mindlin and Cundall–Strack granular packings," *Computers and Geotechnics*, vol. 200, Article 108453, 2026.
   DOI: 10.1016/j.compgeo.2026.108453 | Article.

3. **Pramay Bhatpahri** and Srinivasan Gopalakrishnan, "Prediction of Small-Strain Properties of Dry Sand Using Curve Fitting and Machine Learning Models," *EPJ Web of Conferences*, vol. 340, Article 09013, 2025.
   DOI: 10.1051/epjconf/202534009013 | Article | PDF.

4. **Bhatpahri Pramay**, S. Gopalakrishnan, and Anay Mohan Shembekar, "Constitutive Modelling of Granular Materials Using Cundall–Strack and Hertz–Mindlin Contact Models," *Proceedings of the ASME 2025 Aerospace Structures, Structural Dynamics, and Materials Conference*, Houston, TX, USA, 5–7 May 2025. Paper SSDM2025-152166 / V001T03A016.
   DOI: 10.1115/SSDM2025-152166 | Article | PDF.

Manuscript Under Review

5. **B. Pramay** and S. Gopalakrishnan, "Breakage-Induced Shifts in Pressure Exponents, Void-Ratio Factors, and Separability of Small-Strain Elastic Moduli in Granular Materials." Manuscript under review.

Manuscript in Preparation

6. **B. Pramay** et al., "Contact-Law-Dependent Coherence of Small-Strain Fabric and Force Anisotropies in DEM Sphere Packings: Descriptor-Family Partitioning Distinguishes Hertz–Mindlin from Cundall–Strack Packings." Manuscript in preparation.

Conference Presentations

- **Powders & Grains 2025.** Presenter of published conference paper 3.
- **ASME SSDM 2025.** Presenter of published conference paper 4; Houston, TX, USA, 5–7 May 2025.

Academic and Engineering Experience

Faculty Member Aug 2016 – Jul 2019
Faculty of Engineering and Technology, ICFAI University, Raipur

- Taught undergraduate engineering courses, supervised B.Tech. projects and laboratory sessions, and mentored students in design and simulation.
- Contributed to curriculum development, examinations and evaluation, laboratory coordination, and departmental academic activities.

M.Tech. Thesis Research 2015 – 2016
NIT Raipur / Tata Steel R&D, Jamshedpur

- Characterised API X65 line-pipe steel through tensile, high-cycle fatigue, J-integral fracture-toughness, Charpy impact, grain-size, and microhardness testing, supported by Abaqus analysis.
- Used Leica microscopy, Leco hardness testing, Instron tensile/fatigue systems, and Olympus BX41M / Leica M165 C for fractography.

Industrial Trainee Jun – Jul 2012
Hasdeo Thermal Power Station, Korba, Chhattisgarh
Studied boiler maintenance, coal-based power generation, plant efficiency, and operations.

Industrial Trainee May – Jun 2011
Bhilai Steel Plant, Bhilai, Chhattisgarh
Observed billet and bloom mill operations and studied the process flow from ingot to billet and bloom production.

Engineering Projects

- CAD modelling and analysis of radial-engine components using CATIA V5.
- Stress analysis of a multilayer cylinder using SOLIDWORKS and ANSYS.
- Structural analysis of a beam using SOLIDWORKS and ANSYS.

Certifications, Workshops and Training

- **Computational Engineering & Simulations using Python: FEM & FVM.** Instructor-led online certificate course, iHUB DivyaSampark, IIT Roorkee and RBPL, 1 Jun – 14 Aug 2026.
- Workshop on MEMS, CAD Tools and Smart Structures, Jan 2015.
- C Language Certificate Course, Simply C, Grade A, May – Jun 2013.
- Business Skill Development Programme, Mar 2012.
- AutoCAD Certificate Course, NIT Raipur, Grade A, Aug – Sep 2011.

Academic Awards

- Branch Topper, B.E. Mechanical Engineering, CCEM, Raipur.
- First-year Topper, Mechanical Engineering.

References

Prof. S. Gopalakrishnan
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Professor
Department of Mechanical Engineering
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